Commun. Korean Math. Soc. **20** (2005), No. 4, pp. 685–693

ON A GENERALIZED APERIODIC PERFECT MAP

Sang-Mok Kim

Abstract. An aperiodic perfect map(APM) is an array with the property that every array of certain size, called a window, arises exactly once as a contiguous subarray in the array. In this article, we deal with the generalization of APM in higher dimensional arrays. First, we reframe all known definitions onto the generalized n-dimensional arrays. Next, some elementary known results on arrays are generalized to propositions on n-dimensional arrays. Finally, with some devised integer representations, two constructions of infinite family of n-dimensional APMs are generalized from known 2-dimensional constructions in [7].

1. Introduction

The perfect window property is defined as follows; in a given c-ary $m \times n$ array, every c-ary $u \times v$ array occurs exactly once as a contiguous subarray called a window for fixed positive integers u and v with $u \leq m$ and $v \leq n$. An array satisfying the window property is called an aperiodic perfect map(APM) . If the window property holds in a single period of the array, it is called a periodic perfect map(PM).

The perfect window property has been studied firstly in sequences ([1])-a de Bruijn sequence is a c-ary sequence of period n with the window property that for a given positive integer v with $v \leq n$ every possible c-ary v-tuple occurs in a period of the sequence. Not only many construction methods for de Bruijn sequences have been devised ([4]), but also their existence has been settled ([1, 3, 11]).

For the case of array, many results on PM has been known using the theory of finite field. K. Paterson[9, 10] shows that a PM exists for

Received April 29, 2005.

2000 Mathematics Subject Classification: 94A55, 05B30.

Key words and phrases: de Bruijn sequence, aperiodic perfect map, window property.

The present Research has been conducted by the Research Grant of Kwangwoon University in 2005.

each parameter set satisfying the necessary condition for the existence for any alphabet size c of prime power. G. Hurlbert, C. Mitchell, and K. Paterson[5] show that for each possible parameter set there exists a PM with 2×2 windows.

In contrast to the rich theory of PM, not many results of APMs are known in spite of the practical importance of APM for use on encoding schemes for a two dimensional position sending ([2]). For a given PM, an APM of slightly larger size can be obtained by a simple extension which is called the *closure* of the PM (S.Kanetkar and M.Wagh[6]). However, there are still sets of parameters left satisfying necessary conditions for the existence of APM if the array is "long and thin" enough or its alphabet size is not a prime power ([7]). The first result on the existence of APM is given by C. Mitchell only for the binary case with 2×2 windows. In 2002, the complete answer for the existence of APM for 2×2 windows is given by S. Kim[7] for any alphabet size c.

In this article, we generalize the theory of APM to the higher dimensional array. The generalized higher dimensional window property has not been studied while there could be an analogous application to communication such as determining positions in a cuboid with a device which can examine a certain size of subcuboid ([2]). We firstly reframe all known definitions onto the generalized n-dimensional arrays. Next, we generalize some elementary results on arrays to n-dimensional arrays. Finally, two constructions for infinite family of generalized APMs are given, which is the generalizations of constructions given by S. Kim[7]. We devise some representations of integers in terms of linear combination of consecutive multiples of given integers to prove the constructions being APMs.

2. Generalized definitions and preliminary results

For given positive integers m_j with $j = 1, ..., n$, let I_j be an m_j-set and let C be a non-empty finite set with $|C| = c$. A function $a : I_1 \times ... \times I_n \to C$ is called a c-ary $m_1 \times \ ... \ \times m_n$ array. Usually, we assume that $I_j = \{0, 1, \ ... \ , m_j - 1\}$ and the function a corresponds to an array $A = (a_{i_1 \ ... \ i_n})$, where

$$a_{i_1 \ ... \ i_n} = a\,(i_1, \ ... \ , i_n)$$

for all $(i_1, \ ... \ , i_n) \in I_1 \times \ ... \ \times I_n$. The domain of the function a is called the *index set* and the codomain C is called the alphabet set.

A is called an n-dimensional c-ary $m_1 \times \ ... \ \times m_n$ array if A is c-ary $m_1 \times \ ... \ \times m_n$ array such that $m_j \geq 2$ for all $j = 1, ..., n$.

Let $A = (a_{i_1 \ ... \ i_n})$ be an n-dim c-ary $m_1 \times \ ... \ \times m_n$ array and let u_i be a positive integer with $u_i < m_i$ for $i = 1, ..., n$. For $(s_1, s_2, ..., s_n) \in I_1 \times \ ... \ \times I_n$, the $(s_1, ..., s_n)$-th $u_1 \times ... \times u_n$ window of A is a $u_1 \times ... \times u_n$ array $(\alpha_{i_1 \ ... \ i_n})$ written as $[a_{s_1 \ ... \ s_n}]_{u_1 \times ... \times u_n}$ to be defined as

$$\alpha_{i_1 \ ... \ i_n} = a_{j_1 \ ... \ j_n},$$

where $j_k \equiv i_k + s_k (mod\ m_k)$ for $0 \leq i_k \leq u_k - 1$, $k = 1, 2, ..., n$. For given integers γ_k with $1 \leq \gamma_k \leq m_k$ $(k = 1, 2, ..., n)$, we define

$$[A_{\gamma_1 \times ... \times \gamma_n}]_{u_1 \times ... \times u_n} = \{[a_{s_1 \ ... \ s_n}]_{u_1 \times ... \times u_n} \mid 0 \leq s_k \leq \gamma_k - 1, \ 1 \leq k \leq n\}$$

which is called the set of $u_1 \times ... \times u_n$ *windows* for $\gamma_1 \times ... \times \gamma_n$. If $\gamma_k = m_k$ for $k = 1, ..., n$, we denote $[A_{\gamma_1 \times ... \times \gamma_n}]_{u_1 \times ... \times u_n}$ by $[A]_{u_1 \times ... \times u_n}$. When the size of window is given, it will be simply written as $[A_{\gamma_1 \times ... \times \gamma_n}]$.

Definition 1. Let c be a positive integer with $c \geq 2$. For $i = 1, 2, \ ... \ n$, let m_i and u_i be positive integers satisfying $m_i \geq 2$. An n-dimensional c-ary $m_1 \times ... \times m_n$ array $A = (a_{i_1 \ ... \ i_n})$ is called a $c - ary\ (m_1, ..., m_n; u_1, ..., u_n)$ n-dimensional Periodic Perfect Map (or simply PM_n) if every c-ary $u_1 \times ... \times u_n$ array occurs exactly once as a window $[a_{s_1 \ ... \ s_n}]_{u_1 \times \ ... \ \times u_n}$, where s_i satisfies $0 \leq s_i \leq m_i - 1$ for $i = 1, 2, ..., n$.

The term "periodic" comes from the way that the windows "wrap-round" the array due to the modulo arithmetic on the indices in Definition 1. We immediately have the following lemma as necessary conditions for the parameters of an PM_n.

Lemma 2. *If A is an $c - ary\ (m_1, ..., m_n; u_1, ..., u_n) PM_n$, then the parameters satisfy the following:*

(i) $m_i > u_i$ *for each* $i = 1, 2, ..., n$.
(ii) $m_1 m_2 ... m_n = c^{u_1 u_2 ... u_n}$.

If the positive integers c, m_i and u_i satisfy the necessary conditions for existence of a PM_n in Lemma 2, the set of ordered integers $(c; m_1, ..., m_n; u_1, ..., u_n)$ are called an *admissible parameter set for* PM_n.

Note that the $c^{u_1 u_2 ... u_n}$ windows $[a_{s_1 \ ... \ s_n}]_{u_1 \times ... \times u_n}$ $(0 \leq s_i \leq m_i - 1$, $i = 1, ..., n)$ of a c-ary $(m_1, ..., m_n; u_1, ... \ , u_n)\mathrm{PM}_n$ are distinct. We call a 2-dimensional PM_2 just a Periodic Perfect Map (written as PM).

Next, we define an n-dimensional Aperiodic Perfect Map. In this case, the window does not wrap-round in any index.

DEFINITION 3. Let c be a positive integer with $c \geq 2$. For $i = 1, \ldots, n$, let m_i and u_i be positive integers satisfying $m_i \geq 2$. A n-dimensional c-ary $m_1 \times ... \times m_n$ array $A = (a_{i_1 \ldots i_n})$ is called a $c-ary$ $(m_1, ..., m_n; u_1, ..., u_n)$ n-dimensional Aperiodic Perfect Map (or simply APM_n) if every c-ary $u_1 \times ... \times u_n$ array occurs exactly once as a window $[a_{s_1 \ldots s_n}]_{u_1 \times \ldots \times u_n}$ of A with $0 \leq s_i \leq m_i - u_i$ for $i = 1, ..., n$.

The following necessary conditions for the existence of an APM_n are immediate from those of APM stated in [8].

LEMMA 4. *If A is a $c-ary$ $(m_1, ..., m_n; u_1, ..., u_n)APM_n$, then the parameters satisfy the following:*

(i) $m_i \geq u_i$ *for each* $i = 1, \ldots, n$,

(ii) $\prod_{i=1}^{n}(m_i - u_i + 1) = c^{u_1 u_2 \ldots u_n}$.

We call $(c; m_1, m_2, ..., m_n; u_1, u_2, ..., u_n)$ an *admissible parameter set* for APM_n if it satisfies the conditions given in the above lemma. Note that the simplified notation a $c-ary$ $(m, n; u, v)$APM means the $c-ary$ $(m_1, m_2; u_1, u_2)\mathrm{APM}_2$ where $m_1 = m$, $m_2 = n$, $u_1 = u$ and $u_2 = v$. We remark that, without loss of generality, we restrict our definition of admissible parameters sets for $c-ary$ $(m_1, ..., m_n; u_1, ..., u_n)\mathrm{PM}_n$ (or APM_n) under the condition that $m_1 \leq m_2 \leq ... \leq m_n$ (see [7]).

Next, we describe a construction of an APM_n from the simple extension of a PM_n which extends the definition of the closure of PM given in [7].

DEFINITION 5. Let $A = (a_{i_1 \ldots i_n})$ be a $c-ary$ $(m_1, ..., m_n; u_1, ..., u_n)$ PM_n. We define a n-dim c-ary $(m_1 + u_1 - 1) \times ... \times (m_n + u_n - 1)$ array $\bar{A} = (\bar{a}_{i_1 \ldots i_n})$ by

$$\bar{a}_{i_1 \ldots i_n} = a_{s_1 \ldots s_n}$$

for $0 \leq i_k \leq m_k + u_k - 1$, $k = 1, 2, ..., n$, where $i_k \equiv s_k \pmod{m_k}$. $\bar{A}$ is called the closure of A.

Now we state the following lemma which is generalized from the result due to S. Kanetkar and M. Wage[6].

LEMMA 6. *If A is a $c-ary$ $(m_1, ..., m_n; u_1, ..., u_n)PM_n$ then $\bar{A}$ is a $c-ary$ $(m_1 + u_1 - 1, ..., m_n + u_n - 1; u_1, ..., u_n)APM_n$.*

PROOF. Let A be a $c-ary$ $(m_1, ..., m_n; u_1, ..., u_n)\mathrm{PM}_n$. It is clear that

$$[A_{m_1 \times \ldots \times m_n}]_{u_1 \times \ldots \times u_n} = \left[\bar{A}_{m_1 \times \ldots \times m_n}\right]_{u_1 \times \ldots \times u_n}.$$

From the definition of PM_n and APM_n, we have $\bar{A}$ is a $c-ary$ $(m_1 + u_1 - 1, ..., m_n + u_n - 1; u_1, ..., u_n)\mathrm{APM}_n$. □

REMARK 7. Note that a $c-ary$ $(m_1, ..., m_n; u_1, ..., u_n)\mathrm{APM}_n$ arises by this construction from a $c-ary$ $(m_1-u_1+1, ..., m_n-u_n+1; u_1, u_2, ..., u_n)$ PM_n. The latter must satisfies the conditions that $u_k = m_k - u_k + 1 = 1$ or $m_k - u_k + 1 > u_k$ for $k = 1, 2, ..., n$, i.e., $m_k = u_k = 1$ or $m_k \geq 2u_k$ for $k = 1, 2, ..., n$. Since a PM_n is an n-dim array, so $m_k > 1$ so that the parameters of the APM_n necessary satisfy $m_k \geq 2u_k$ for all $k = 1, 2, ..., n$.

3. Some generalized APM_ns from APM_2s in [7]

In this section two constructions of APMn is given from APM_2 which are devised to give a complete answer for the existence of APM for 2×2 windows (see S. Kim[7]). The one is a c-ary $(u_1, ..., u_{n-1}, m_n; u_1, ..., u_{n-1}, u_n)$APM, and the other is a c-ary $(2u_1 - 1, ..., 2u_{n-1} - 1, m_n\ ;\ u_1, ..., u_n)$ APM.

We first observe some properties on integers which immediately follow from representation of integers.

REMARK 8. For $1 \leq i \leq n$, let u_i be a positive integer with $2 \leq u_i$ and let c be any positive integer.

(1) Let $z = \prod_{s=1}^{n} u_s$ and let b be an integer with $0 \leq b \leq c^z - 1$. If we consider b as an integer in base c, then b may be uniquely written as
$$b = \sum_{s=0}^{z-1} b_s c^s,$$
where $0 \leq b_s \leq c - 1$ for each $s = 0, 1, ..., z - 1$.

(2) If t is a non-negative integer such that $0 \leq t \leq (\prod_{s=1}^{n} u_s) - 1$, then t may be written uniquely as
$$t = \pi_1(t) + \pi_2(t)u_1 + \pi_3(t)u_1u_2 + ... + \pi_n(t)u_1u_2...u_{n-1},$$
where $\pi_i(t)$ is an integer with $0 \leq \pi_i(t) \leq u_i - 1$ for $i = 1, 2, ..., n$.

Now we generalize the construction of c-ary $(u, n; u, v)$APM in [7] to the n-dimensional case.

CONSTRUCTION 9. *c-ary $(u_1, ..., u_{n-1}, m_n\ ; u_1, ..., u_{n-1}, u_n)APM_n$. Let $(c; u_1, ..., u_{n-1}, m_n; u_1, ..., u_{n-1}, u_n)$ be an admissible parameter set for APM_n. Note that from Lemma 4, we have*
$$m_n = c^{u_1u_2...u_n} + u_n - 1.$$

(1) *Let* $B = (b_s)$ *be a* $c^{u_1u_2\ldots u_{n-1}}$*-ary span* u_n *de Bruijn sequence and* $\bar{B} = \left(\bar{b}_t\right)$ *be the closure of* B*. i.e.,* $\bar{B} = \left(\bar{b}_t\right)$ *is defined by* $\bar{b}_t = b_s$ *if and only if* $t = s \pmod{c^{u_1u_2\ldots u_n}}$ *and it is a 1-dim* $c^{u_1u_2\ldots u_{n-1}}$*-ary* m_i *array which is a* $c^{u_1u_2\ldots u_{n-1}}$*-ary* $(c^{u_1u_2\ldots u_n} + u_n - 1; u_n)APM_1$.

(2) *Let* $t = s \pmod{c^{u_1u_2\ldots u_n}}$*. For any* $\bar{b}_t = b_s$*, from Remark 8,*

(i) b_s *can be written as*

$$b_s = \sum_{i=0}^{u_1u_2\ldots u_{n-1}-1} (b_s)_i c^i,$$

where $0 \le (b_s)_i \le c - 1$ *for all* $i \in \mathbb{N}_{u_1u_2\ldots u_{n-1}}$.

(ii) *Each* $i \in \mathbb{N}_{u_1u_2\ldots u_{n-1}}$ *may be uniquely written as*

$$i = i_1 + i_2u_1 + i_3u_1u_2 + \ldots + i_{n-1}u_1u_2\ldots u_{n-2},$$

where $0 \le i_t \le u_t - 1$ *for* $t = 1, \ldots, n-1$.

Hence, we write $(b_s)_i = (b_s)_{i_1+i_2u_1+i_3u_1u_2+\ldots+i_{n-1}u_1u_2\ldots u_{n-2}}$.

(3) *Define a n-dim c-ary* $u_1 \times \ldots \times u_{n-1} \times m_n$ *array* $A = \left(a_{i_1\ i_2\ \ldots\ i_{n-1}\ i_n}\right)$ *by*

$$a_{i_1\ i_2\ \ldots\ i_{n-1}\ i_n} = (b_{i_n})_{i_1+i_2u_1+i_3u_1u_2+\ldots+i_{n-1}u_1u_2\ldots u_{n-2}},$$

where $0 \le i_t \le u_t - 1$ *for* $t = 1, \ldots, n-1$ *and* $0 \le i_n \le m_n = c^{u_1u_2\ldots u_n} + u_n - 2$.

THEOREM 10. *Let* $(c; u_1, \ldots, u_{n-1}, m_n\ ;\ u_1, \ldots, u_{n-1}, u_n)$ *be an admissible parameter set for* APM_n*. Let* B *be a* $c^{u_1u_2\ldots u_{n-1}}$*-ary span* u_n *de Bruijn sequence. Then the n-dim* $u_1 \times \ldots \times u_{n-1} \times (c^{uv} + v - 1)$ *array constructed obtained from Construction 9 is a c-ary* $(u_1, \ldots, u_{n-1}, m_n; u_1, \ldots, u_{n-1}, u_n)APM_n$.

PROOF. We prove that $A = \left(a_{i_1\ i_2\ \ldots\ i_{n-1}\ i_n}\right)$ in Construction 9 is a c-ary $(u_1, \ldots, u_{n-1}, m_n; u_1, \ldots, u_n)$ APM_n. Note that since $m_i = u_i$ for $i = 1, \ldots, n-1$ and $m_n = c^{u_1u_2\ldots u_n} + u_n - 1$, any $u_1 \times \ldots \times u_n$ window $[a_{k_1\ k_2\ \ldots\ k_n}]$ in $[A_{1\times 1\times\ldots\times 1\times(m_n-u_n+1)}]$ has $k_j = 0$ for $j = 1, \ldots, n-1$ and $0 \le k_n \le c^{u_1u_2\ldots u_n} - 1$. Let $[a_{0\,0\,\ldots\,0\,l}] = (\alpha_{s_1\ \ldots\ s_n})$ and $[a_{0\,0\,\ldots\,0\,k}] = (\beta_{s_1\ \ldots\ s_n})$ be any two $u_1 \times \ldots \times u_n$ windows in $[A_{1\times 1\times\ldots\times 1\times(m_n-u_n+1)}]$, where $0 \le l, k \le c^{u_1u_2\ldots u_n} - 1$. Suppose that $[a_{0\,0\,\ldots\,0\,l}] = [a_{0\,0\,\ldots\,0\,k}]$. Then, for all $s_t = 0, 1, \ldots, u_t - 1$, $t = 0, 1, \ldots, n$, we have

$$\alpha_{s_1\ \ldots\ s_n} = a_{s_1\ \ldots\ s_{n-1}\ s_n+l} = a_{s_1\ \ldots\ s_{n-1}\ s_n+k} = \beta_{s_1\ \ldots\ s_n}.$$

From Construction 9, it implies

$$\begin{aligned}&(b_{s_n+l})_{s_1+s_2u_1+s_3u_1u_2+\ldots+s_{n-1}u_1u_2\ldots u_{n-2}}\\ &= (b_{s_n+k})_{s_1+s_2u_1+s_3u_1u_2+\ldots+s_{n-1}u_1u_2\ldots u_{n-2}}.\end{aligned}$$

It immediately follows that for all $b_{l+s_n} = b_{k+s_n}$ for s_n with $0 \leq s_n \leq u_n - 1$, which implies $[b_l]_{u_n} = [b_k]_{u_n}$ in $[B_{c^{u_1u_2\cdots u_{n-1}}}]_{u_n}$. From the definition of de Bruijn sequence, it follows that $l = k$.

Therefore, $\left|[A_{1\times\ldots\times 1\times(m_n-u_n+1)}]\right| = c^{u_1u_2\ldots u_n}$ which implies that each c-ary $u_1 \times \ldots \times u_n$ array occurs exactly once in $[A_{1\times 1\times\ldots\times 1\times(m_n-u_n+1)}]$ and A is a c-ary $(u_1, \ldots, u_{n-1}, m_n; u_1, \ldots, u_n)\mathrm{APM}_n$. □

Next, we generalize the construction of c-ary $(2u-1, n; u, v)\mathrm{APM}_2$ given in [7] to the n-dimensional case.

Construction 11. *c-ary $(2u_1 - 1, \ldots, 2u_{n-1} - 1, m_n; u_1, \ldots, u_{n-1}, u_n)APM_n$. Let $(c; 2u_1-1, \ldots, 2u_{n-1}-1, m_n; u_1, \ldots, u_n)$ be an admissible parameter set for APM_n. Note that, from Lemma 4, we have*

$$m_n = \frac{c^{u_1u_2\ldots u_n}}{u_1u_2\ldots u_{n-1}} + u_n - 1$$

and so $c^{u_1u_2\ldots u_n}$ is divisible by $u_1u_2\ldots u_{n-1}$. Let $B = (b_i)$ be a c-ary span $u_1u_2\ldots u_n$ de Bruijn sequence and let $\bar{B} = (\bar{b}_i)$ be the closure of B which is a c-ary $(1, c^{u_1u_2\ldots u_n} + u_1u_2\ldots u_n - 1; 1, u_1u_2\ldots u_n)APM$. Define a n-dim $(2u_1 - 1) \times \ldots \times (2u_{n-1} - 1) \times m_n$ array $A = (a_{i_1\,\ldots\,i_n})$ by

$$a_{i_1\,i_2\,\ldots\,i_n} = \bar{b}_{i_1+i_2u_1+i_3u_1u_2+\ldots+i_nu_1u_2\ldots u_{n-1}}.$$

for $0 \leq i_j \leq 2u_j - 2$, $j = 1, \ldots, n-1$ and $0 \leq u_j \leq m_n - 1 = \frac{c^{u_1u_2\ldots u_n}}{u_1u_2\ldots u_{n-1}} + u_n - 2$. Note that $\bar{b}_k = b_l$ for $l \equiv k \pmod{c^{u_1u_2\ldots u_n}}$.

Theorem 12. *Let B be a c-ary span $u_1u_2\ldots u_n$ de Bruijn sequence. Let $(c; 2u_1-1, \ldots, 2u_{n-1}-1, m_n; u_1, \ldots, u_n)$ be an admissible parameter set for APM_n. Then the n-dim $(2u_1 - 1) \times \ldots \times (2u_{n-1} - 1) \times m_n$ array constructed from B by Construction 11 is a c-ary $(2u_1-1, \ldots, 2u_{n-1}-1, m_n; u_1, \ldots, u_n)APM_n$.*

Proof. Let $(c; 2u_1-1, \ldots, 2u_{n-1}-1, m_n; u_1, \ldots, u_n)$ be an admissible parameter set for APM_n. Let $A = (a_{i_1\,\ldots\,i_n})$ be an array constructed as in Construction 11 from a $c - ary$ span $u_1u_2\ldots u_n$ de Bruijn sequence $B = (b_i)$. Then, from Lemma 4, we have

$$m_n = \frac{c^{u_1u_2\ldots u_n}}{u_1u_2\ldots u_{n-1}} + u_n - 1$$

and $c^{u_1u_2\ldots u_n}$ is divisible by $u_1u_2\ldots u_{n-1}$. Let $\bar{B} = \left(\bar{b}_i\right)$ be the closure of B. For each i_k with $0 \leq i_k \leq u_k - 1$ $(k = 1, \ldots, n)$, let $[a_{s_1\,s_2\,\ldots\,s_n}] = (\alpha_{i_1\,i_2\,\ldots\,i_n})$ and $[a_{t_1\,t_2\,\ldots\,t_n}] = (\beta_{i_1\,i_2\,\ldots\,i_n})$ be $u_1 \times \ldots \times u_n$ windows in

$$[A_{u_1\times\,\ldots\,\times u_{n-1}\times(m_n-u_n+1)}]_{u_1\times\ldots\times u_n}.$$

Note that $0 \le s_i, t_i \le u_i - 1$ for $i = 1, \ \dots \ , n-1$ and $0 \le s_n, t_n \le m_n - u_n = \frac{c^{u_1 u_2 \dots u_n}}{u_1 u_2 \dots u_{n-1}} - 1$.

Suppose that $[a_{s_1\ s_2\ \dots\ s_n}] = [a_{t_1\ t_2\ \dots\ t_n}]$. Then, for $0 \le i_k \le u_k - 1$ and $k = 1, \dots, n$, we have $\alpha_{i_1\ i_2\ \dots\ i_n} = \beta_{i_1\ i_2\ \dots\ i_n}$ so that

$$\begin{aligned}
\alpha_{i_1\ i_2\ \dots\ i_n} &= a_{s_1+i_1\ s_2+i_2\ \dots\ s_n+i_n} \\
&= \bar{b}_{(s_1+i_1)+(s_2+i_2)u_1+(s_3+i_3)u_1u_2+\dots+(s_n+i_n)u_1u_2\dots u_{n-1}} \\
&= \bar{b}_{(t_1+i_1)+(t_2+i_2)u_1+(t_3+i_3)u_1u_2+\dots+(t_n+i_n)u_1u_2\dots u_{n-1}} \\
&= a_{t_1+i_1\ t_2+i_2\ \dots\ t_n+i_n} \\
&= \beta_{i_1\ i_2\ \dots\ i_n}.
\end{aligned}$$

Thus, we have

$$\begin{aligned}
&\bar{b}_{(i_1+i_2u_1+i_3u_1u_2+\dots+i_nu_1u_2\dots u_{n-1})+(s_1+s_2u_1+s_3u_1u_2+\dots+s_nu_1u_2\dots u_{n-1})} \\
&= \bar{b}_{(i_1+i_2u_1+i_3u_1u_2+\dots+i_nu_1u_2\dots u_{n-1})+(t_1+t_2u_1+t_3u_1u_2+\dots+t_nu_1u_2\dots u_{n-1})}.
\end{aligned}$$

Let $l = i_1 + i_2u_1 + i_3u_1u_2 + \dots + i_nu_1u_2\dots u_{n-1}$. Then, for all l with $0 \le l \le u_1u_2\dots u_n - 1$, we have

$$\bar{b}_{l+(s_1+s_2u_1+s_3u_1u_2+\dots+s_nu_1u_2\dots u_{n-1})} = \bar{b}_{l+(t_1+t_2u_1+t_3u_1u_2+\dots+t_nu_1u_2\dots u_{n-1})}$$

which implies that

$$\begin{aligned}
&[b_{s_1+s_2u_1+s_3u_1u_2+\dots+s_nu_1u_2\dots u_{n-1}}]_{u_1u_2\dots u_n} \\
&= [b_{t_1+t_2u_1+t_3u_1u_2+\dots+t_nu_1u_2\dots u_{n-1}}]_{u_1u_2\dots u_n}.
\end{aligned}$$

From the definition of a de Bruijn sequence B, since $0 \le s_k, t_k \le u_k - 1$ when $1 \le k \le n-1$ and $0 \le s_n, t_n \le c^{u_1u_2\dots u_n} - 1$, we have

$$\begin{aligned}
&s_1 + s_2u_1 + s_3u_1u_2 + \dots + s_nu_1u_2\dots u_{n-1} \\
&= t_1 + t_2u_1 + t_3u_1u_2 + \dots + t_nu_1u_2\dots u_{n-1}.
\end{aligned}$$

By Remark 8, we have $s_i = t_i$ for all $i = 1, \ \dots \ , 0$.

Therefore, we conclude that $[a_{s_1\ s_2\ \dots\ s_n}] = [a_{t_1\ t_2\ \dots\ t_n}]$ if and only if $s_i = t_i$ for all $i = 1, \ \dots \ , 0$. Thus, we have

$$\left|[A_{(m_1-u_1+1)\times\dots\times(m_n-u_n+1)}]\right| = \left|[A_{u_1\times\ \dots\ \times u_{n-1}\times(\frac{c^{u_1u_2\dots u_n}}{u_1u_2\dots u_{n-1}})}]\right| = c^{u_1u_2\dots u_n}$$

so that every c-ary $u_1 \times \dots \times u_n$ array occurs exactly once as a window i.e., A is a c-ary $(2u_1 - 1, \ \dots \ , 2u_{n-1} - 1, m_n; u_1, \ \dots \ , u_n)\mathrm{APM}_n$. □

References

[1] N. de Bruijn, *A combinatorial problem*, Proc. Nederlandse Akademie van Wetenschappen **49** (1946), 758–764.

[2] J. Burns and C. Mitchell, *Coding Schemes for two-dimensional position sending*, Cryptography and Coding III, M. Ganley Ed. London, UK: Oxford Univ. Press, pp.31–61, 1993.

[3] I. J. Good, *Normally recurring decimals*, J. London Math. Soc. **21** (1946), 167–169.

[4] H. Fredricksen, *A survey of full length nonlinear shift register cycle algorithms*, SIAM J. Algebraic and Discrete Methods **1** (1980), 107–113.

[5] G. Hurlbert, C. J. Mitchell, and K. G. Paterson, *On the existence of de Bruijn tori with two by two windows*, J. Combin. Theory Ser. A **76** (1996), 213–230.

[6] S. Kanetkar and M. Wagh, *On Construction of Matrices with distict submatrices*, SIAM J. Algebraic and Discrete Methods **1** (1980), 107–113.

[7] S. Kim, *On the existence of aperiodic perfect maps for 2 by 2 windows*, Ars Combin. **65** (2002), 111–120.

[8] C. J. Mitchell, *Aperiodic and semi-periodic perfect maps*, IEEE Trans. Inform. Theory **41** (1995), 88–95.

[9] K. G. Paterson, *New Classes of Perfect Maps I*, J. Combin. Theory Ser. A **73** (1996), 302–334.

[10] ———, *New Classes of Perfect Maps II*, J. Combin. Theory Ser. A **73** (1996), 335–345.

[11] D. Rees, *Note on a paper by I. J. Good*, J. London Math. Soc. **21** (1946), 169–172.

Division of General Education
Mathematics
Kwangwoon University
Seoul 139-701, Korea
E-mail: smkim@kw.ac.kr